# Using Feature Flags to play around with production code

Feature Flags are nifty at branching functionality within your application to run experiments without affecting functionality of the application

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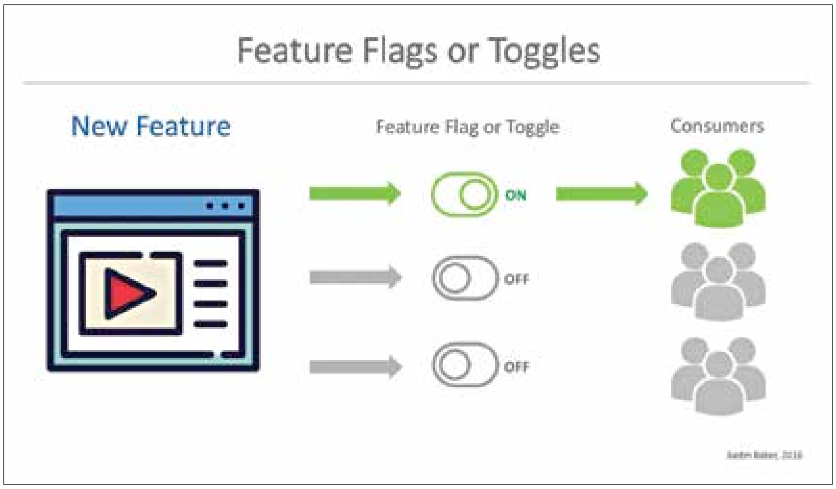


Use feature flags to branch your application

Feature Flags are a great way to introduce new changes to your code base without breaking existing functionality. No matter what kind of software development model you follow, there always exists the need to iteratively improve the functionality of the application. It may be to introduce new features, improve existing ones, remove vulnerabilities or it could be any of the millions of reasons why you want to mess with code. However, this presents the inherent problem of possibly breaking the functionality of your existing application; and if this application happens to be on a production server where it is critical to your service or product then you're playing a dangerous game. This is where you'd want to use Feature Flags to branch your application.

Feature Flags have recently been gaining popularity thanks to the folks over at Facebook, Uber, and Twitter going gaga over it in their blog posts. However, if you look back into the past, Feature Flags have been used by developers for a much longer time. There is a Gmail blog post from 2011 that mentions the use of Feature Flags, so the folks over at Facebook, Twitter and Uber are a little late to the party. At least when you consider when they announced the adoption of the technique via their official blogs.

## HOW DO FEATURE FLAGS WORK

In its simplest form, think of Feature Flags as a simple conditional statement. If the flag is switched on, then one branch of the code gets executed and if the flag is switched off, then another branch of the code gets executed. So you can route all your existing requests/users onto the older but functional code while you tinker with the new code in a different branch. All of this is happening on the production server. Feature flags are also called feature toggles, release toggles or feature switches. Here's an example of how it can work:

**Old Code:**

```
function switchMeOut(){
        // Existing function
used on production
  }
```

**New Code:**

```
function switchMeOut(){
        var useNewFunction =
false;
        // useNewFunction =
true; // Uncomment when you
want to switch to new func-
tion
        if( useNewFunction ){
        return newFunctionRe-
sults();
        }else{
        return oldFunctionRe-
sults();
        }
```